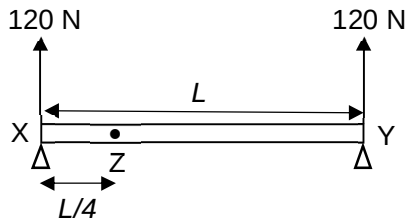


- 7 A uniform plank of length L is supported by initially equal forces of 120 N at X and Y. If the force at X is now moved to Z such that the plank is kept horizontal, what is the magnitude of the force at Y?



A 60 N



B 80 N

C 120 N

D 160 N

Weight of plank is 240 N acting through the centre.

When force is moved to point Z, taking moment about point Z,

$$240 \times L/4 = \text{Force at Y} \times 3L/4$$

$$\text{Force at Y} = 80 \text{ N}$$

8 A ball is projected vertically upwards. At a height h the kinetic energy of the ball is K and its